

Zeyu Chen

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EDUCATION

The University of Hong Kong (HKU)

Ph.D. in Artificial Intelligence

Hong Kong

Sep. 2024 – present

The University of Hong Kong (HKU)

M.S. in Data Science

Hong Kong

Sep. 2022 – Jan. 2024

Nanjing University of Science and Technology

B.S. in Applied Statistics

Nanjing, China

Sep. 2016 – Jun. 2020

EXPERIENCE

Feed-forward 4D Clothed Human Reconstruction

Feb. 2025 – Present

Visual AI Lab, HKU (Advisor: Prof. Kai Han)

Hong Kong

- Leverage 3D Gaussian Splatting to reconstruct high-fidelity 3D human avatars from sparse-view inputs, optimizing for both geometric accuracy and rendering speed.
- Develop a dual-layer decoupling pipeline to separate body and clothing representations, introducing a novel animation scheme via extended Linear Blend Skinning (LBS) to recover dynamic cloth deformations from monocular videos.
- Achieve state-of-the-art (SOTA) performance, outperforming baselines such as *LHM* and *GHG* in terms of reconstruction quality, structural integrity, and visual fidelity.

Multimodal Unified Embedding Space Learning

Jan. 2024 – Sep. 2025

Visual AI Lab, HKU (Advisor: Dr. Jie Li, Prof. Kai Han)

Hong Kong

- Developed CodeBind, a unified framework to align nine distinct modalities (including image, text, audio, video, thermal, and depth) into a joint embedding space via Vector Quantization (VQ) techniques.
- Proposed a decoupled representation learning scheme using modality-shared and modality-specific codebooks, effectively isolating cross-modal invariant features from domain-specific characteristics.
- Surpassed industry-standard baselines (*ImageBind*, *ViT-Lens*) across multiple benchmarks, achieving SOTA results in zero-shot multimodal classification and cross-modal retrieval tasks.
- Conducted in-depth analysis of language density's role in multimodal alignment and leveraged Qwen3-VL to interpret and visualize the semantic properties of decomposed modality-specific components.
- Demonstrated the versatility of the decoupled space by enabling downstream applications such as any-to-image generation.

Compositional Text-to-Image Generation with Diffusion Model

Jun. 2024 – Dec. 2024

Master Thesis, HKU (Advisor: Dr. Jie Li, Prof. Kai Han)

Hong Kong

- Developed a controllable generation framework based on Stable Diffusion to synthesize compositional images featuring multiple personalized concepts and precise layout arrangements.
- Implemented LoRA-based fine-tuning integrated with an auxiliary image encoder to inject reference features into text embeddings, significantly enhancing identity preservation and training efficiency over the *Mix-of-Show* baseline.
- Engineered a novel training-free guidance mechanism by optimizing cross-attention maps during the sampling process, ensuring that the spatial distribution of personalized concepts strictly adheres to user-defined bounding boxes.
- Demonstrated superior generalization in complex multi-concept scenarios, effectively mitigating the "attribute leakage" and "missing objects" issues common in standard T2I models.

Machine Vision to Facial Fatigue with Nonlocal 3D Attention Network

Feb. 2021 – Sep. 2021

Research Intern, BIAI Intelligence Biotech Co., Ltd (Advisor: Dr. Xiaotao Li)

Shenzhen, China

- Curated a real-world facial fatigue dataset by capturing high-resolution video data in naturalistic driving and working scenarios, moving beyond traditional lab-simulated environments.
- Developed a spatiotemporal feature extraction backbone by upgrading 2D architectures to a 3D ResNet integrated with Non-local Self-Attention mechanisms, achieving 90% accuracy on binary fatigue classification.
- Enhanced model interpretability by utilizing Grad-CAM and guided backpropagation to visualize and validate salient physiological fatigue indicators (e.g., eye closure duration, blinking patterns) captured by the network.

PUBLICATIONS

Chen, Z., Li, J., & Han, K. (2026). CodeBind: Decoupled Representation Learning for Multimodal Alignment with Unified Compositional Codebook. In Findings of the Association for Computational Linguistics: ACL 2026. Association for Computational Linguistics.

Chen, Z., Zhang, X., Li, J., Ni, J., Chen, G., Wang, S., ... & Li, X. (2021). Machine vision detection to daily facial fatigue with a nonlocal 3D attention network. arXiv preprint arXiv:2104.10420.

Ding, Z., **Chen, Z.**, Ma, M., & Zhang, Z. (2019). Application of KF improved by Hermite interpolation in target trajectory correction. In AIP Conference Proceedings (Vol. 2185, No. 1, p. 020051). AIP Publishing LLC.

HONORS & AWARDS

- 2023** – Suoxinda Scholarship in Data Science, HKU (HK\$20,000)
- 2023** – Subject Prize Awardees, HKU (HK\$1000 x4 courses)
- 2017-2019** – NJUST Scholarship for Excellent Students

TECHNICAL SKILLS

- Programming:** Python
- Frameworks & Tools:** PyTorch, OpenCV, Git
- Research Areas:** Multimodal Representation Learning, 4D Human Reconstruction, Generative AI
- Languages:** Chinese (Native), English (Fluent)